

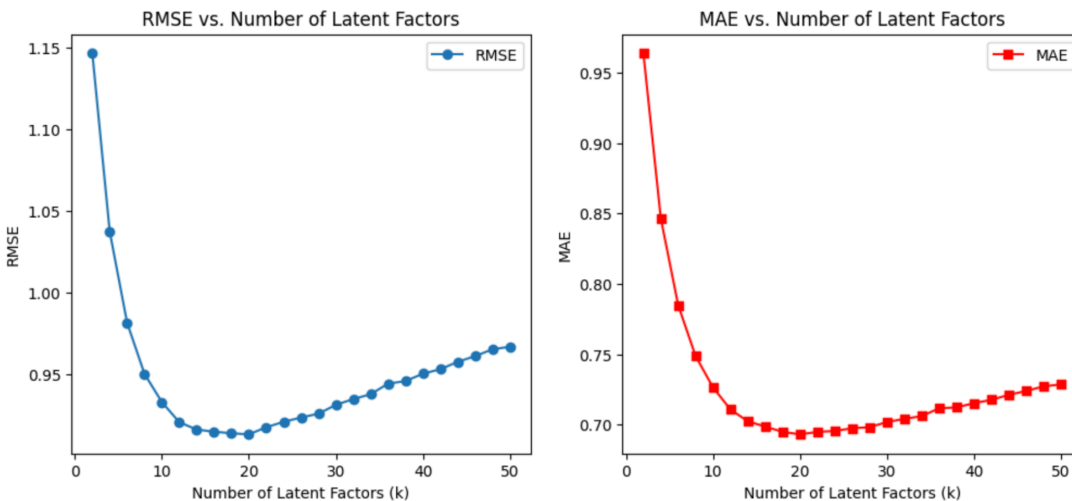
Q7:

No, the problem is non-convex because the objective function involves the product of two optimization variables, U and V, making it bilinear and non-convex.

For fixed U, then the least-square problem will be as follows:

$$\min_V \sum_{i=1}^m \sum_{j=1}^n W_{ij} (r_{ij} - (UV^T)_{ij})^2$$

Q8:



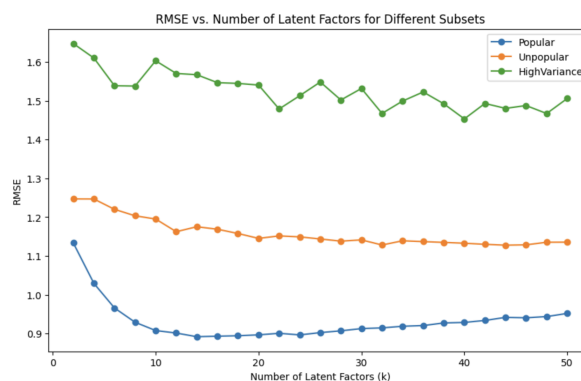
Optimal k (RMSE): 20, Minimum RMSE: 0.9130

Optimal k (MAE): 20, Minimum MAE: 0.6932

Number of unique movie genres: 20

The optimal number of latents is the same as the number of movie genres.

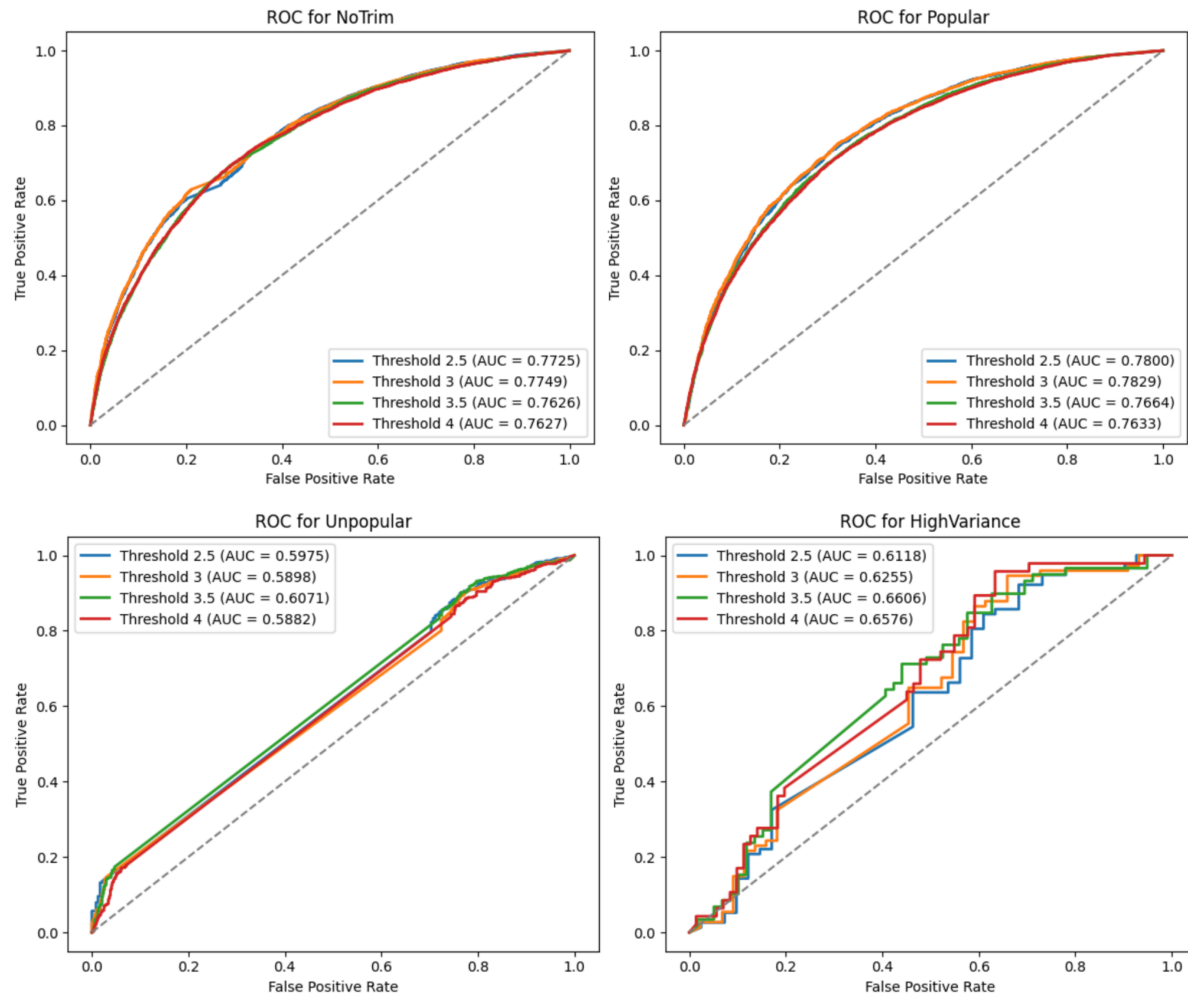
Subsets:



Subset: Popular, Optimal k: 14, Minimum RMSE: 0.8921

Subset: Unpopular, Optimal k: 44, Minimum RMSE: 1.1279

Subset: HighVariance, Optimal k: 40, Minimum RMSE: 1.4533



Q9:

Factor 1:

Fantasy|Horror, Horror|Mystery|Thriller, Horror, Action|Crime|Thriller,
Drama|Thriller, Drama|Horror|Thriller, Drama, Comedy|Drama,
Action|Adventure, Crime|Horror|Mystery

Factor 2:

Children|Comedy, Comedy, Comedy|Fantasy|Horror, Comedy|Romance,
Adventure|Comedy|Sci-Fi, Drama|War, Crime|Drama,
Action|Comedy|Sci-Fi|Thriller, Comedy, Action|Adventure|Children|Comedy

Factor 3:

Comedy|Drama|Romance, Comedy|Crime, Action|Sci-Fi|Thriller, Documentary,
Documentary, Romance|Thriller, Action|Children, Comedy, Comedy|Romance,
Drama|War

Factor 4:

Action|Sci-Fi, Sci-Fi, Horror|Sci-Fi, Comedy|Drama, Comedy,
Action|Drama|Fantasy, Horror|Sci-Fi,

Action|Adventure|Animation|Children|Comedy|Romance, Drama,
Drama|Romance|Western

Factor 5:

Action|Fantasy|Horror|Sci-Fi|Thriller, Comedy|Romance, Drama,
Drama|Horror|Thriller, Action|Drama, Comedy|Fantasy, Thriller,
Horror|Sci-Fi|Thriller, Comedy|Documentary, Comedy|Crime

Factor 6:

Animation|Children|Comedy|Musical,
Adventure|Animation|Children|Comedy|IMAX, Horror|Thriller, Comedy|Romance,
Animation|Children|Comedy, Horror, Action|Comedy|Crime,
Comedy|Drama|Romance, Action|Comedy, Comedy|Drama

Factor 7:

Children|Comedy|Fantasy|Romance, Action|Adventure|Animation,
Action|Comedy|Crime|Fantasy, Comedy|Drama, Children|Comedy|Drama, Comedy,
Action|Comedy|Crime|Thriller, Drama, Action|Comedy, Horror|Mystery

Factor 8:

Action|Adventure|Sci-Fi, Drama|Romance, Documentary, Action|Drama,
Animation|Comedy, Drama|Mystery, Comedy|Fantasy|Horror|Musical|Thriller,
Comedy|Romance, Comedy|Drama|Romance, Drama|Romance

Factor 9:

Action|Horror|Thriller, Children|Comedy, Comedy, Comedy, Comedy,
Drama|Romance, Action|Adventure|Animation|Fantasy,
Fantasy|Romance|Thriller|IMAX, Horror, Adventure|Drama

Factor 10:

Fantasy|Western, Comedy|Romance, Action|Crime|Sci-Fi|Thriller,
Adventure|Children, Action|Adventure|Children|Fantasy, Comedy|Drama,
Comedy, Drama|Mystery|Sci-Fi, Children|Comedy|Fantasy, Comedy|Drama

Analysis:

Factor 1: Mainly includes Horror, Thriller, Drama, Action, Crime, indicating this factor is related to dark, suspenseful, and intense movies.

Factor 2: Mainly includes Comedy, Children, Romance, indicating this factor is related to lighthearted, family-friendly, and humorous movies.

Factor 3: Mainly includes Comedy, Romance, Documentary, Action, indicating this factor is related to entertaining, emotionally rich, or documentary-style movies.

Factor 4: Mainly includes Sci-Fi, Horror, Drama, Comedy, indicating this factor is related to science fiction, futuristic, and thriller movies.

Factor 5: Mainly includes Action, Horror, Thriller, Comedy, Drama, indicating this factor is related to fast-paced, suspenseful, and action-packed movies.

Factor 6: Mainly includes Animation, Children, Comedy, Horror, indicating this factor is related to family-friendly animations and children's movies, with some horror elements.

Factor 7: Mainly includes Children, Comedy, Action, Drama, indicating this factor is related to family-oriented adventure and humorous movies.

Factor 8: Mainly includes Action, Adventure, Sci-Fi, Drama, Romance, indicating this factor is related to adventurous and dramatic storytelling movies.

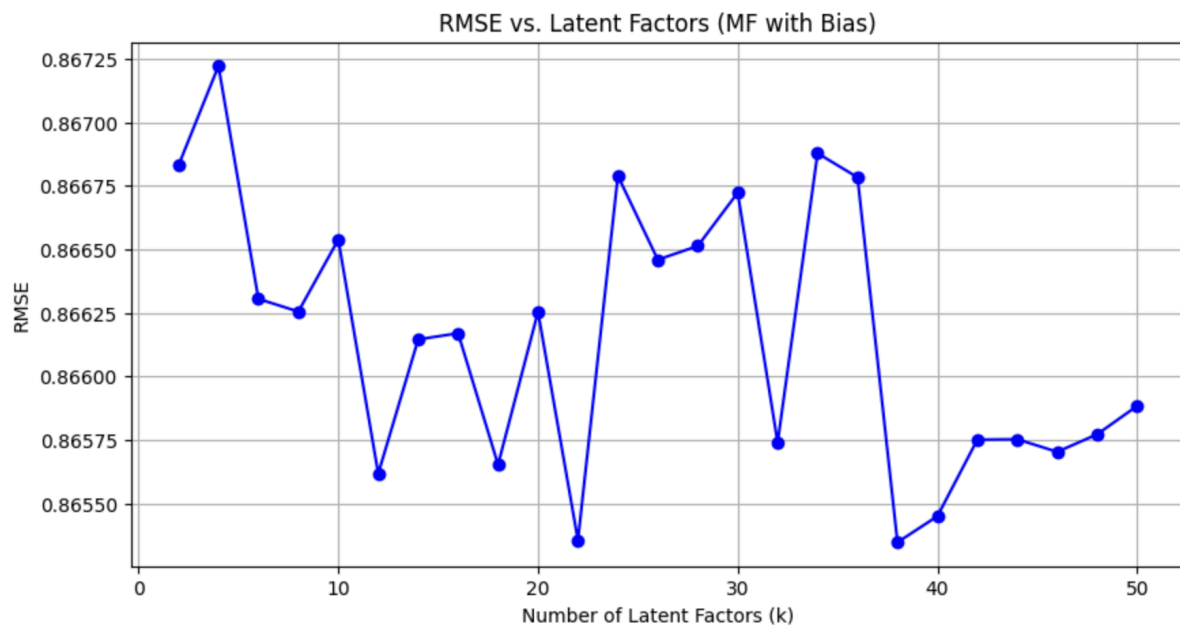
Factor 9: Mainly includes Action, Horror, Thriller, Comedy, Fantasy, indicating this factor is related to high-energy or supernatural-themed movies.

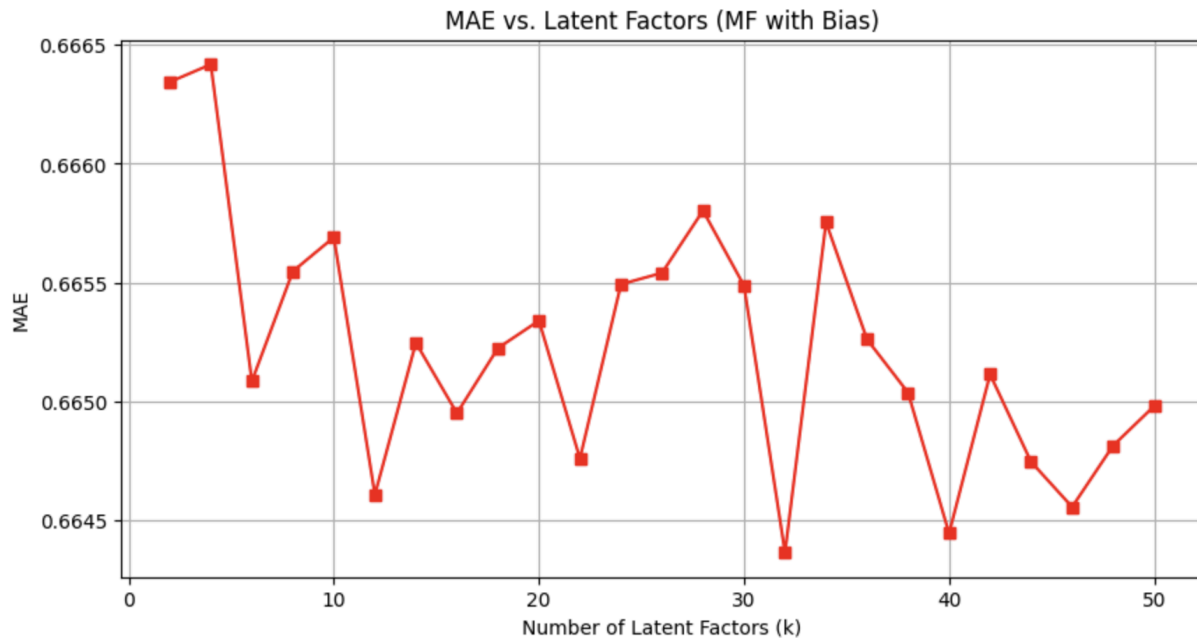
Factor 10: Mainly includes Fantasy, Western, Comedy, Adventure, Sci-Fi, indicating this factor is related to fantasy, science fiction, and western adventure movies.

Conclusion:

1. The top 10 movies in each factor generally belong to a particular or small collection of genres, confirming that the NMF model successfully captures thematic patterns in movies.
1. There is a clear connection between the latent factors and movie genres, with each factor representing a distinct movie style.
2. NMF can be used for genre-based recommendations, grouping users based on their preferences for specific types of movies.

Q10:





```
best_k, best_rmse, best_mae:
(38, 0.8653478210495102, 0.665037894332046)
```

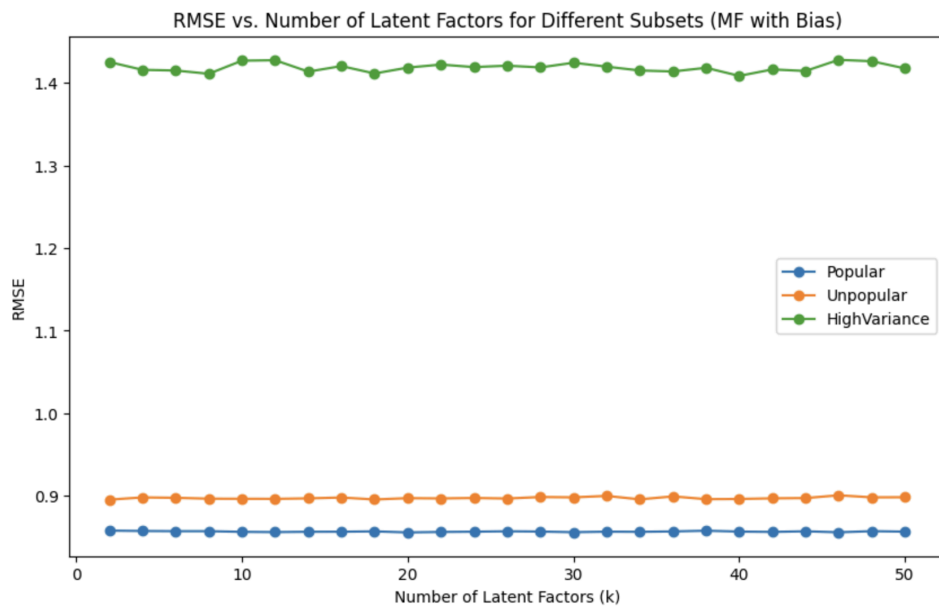
The optimal number 38 is not close to the total number of moviegenres.

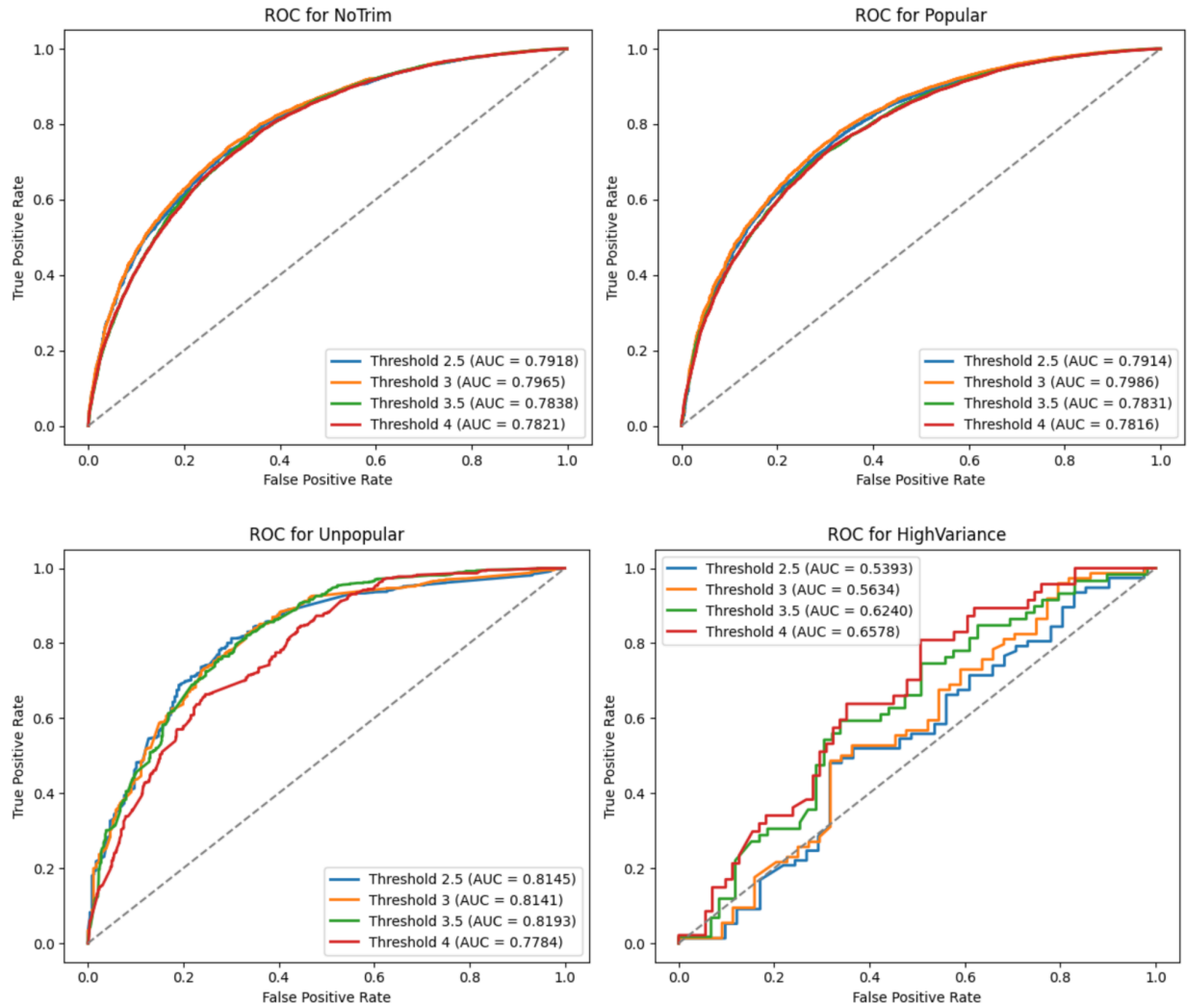
Subsets:

Subset: Popular, Optimal k: 20, Minimum RMSE: 0.8559

Subset: Unpopular, Optimal k: 2, Minimum RMSE: 0.8955

Subset: HighVariance, Optimal k: 40, Minimum RMSE: 1.4084:





Q11

Naïve Collaborative Filter - Average RMSE: 0.9347

Subset: Popular, Average RMSE: 0.9323

Subset: Unpopular, Average RMSE: 0.9710

Subset: HighVariance, Average RMSE: 1.3743

Q12:

The KNN plot in the first figure is incorrect and should be replaced with the one in the second figure with threshold = 3. However, due to our division of tasks and time constraints later on, we were unable to update it to the correct KNN plot in time. Please refer to this and make the necessary adjustments. Thank you.

